



FIELD NOTES • REWARD DESIGN

Paying Agents to Find Truth

How to make a team of AI auditors accurate on purpose — by rewarding provable findings, punishing bluffing, and never letting a model grade its own work.

Give an AI a reason to find problems and it will invent them. This paper documents the incentive design behind the Tiger Team inspection layer: a scoring system engineered so that the only way for an auditor to win points is to be verifiably, citably right — and where a confident lie costs more than silence.

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Series • **Who Audits the Robots?**

Subject • Reward engineering for multi-agent AI evaluation

Companion • the live Tiger Team walkthrough

We call it the *Tiger Team* — a pack of rival AIs whose only job is to hunt one system for flaws, then hand back the fix. A tiger doesn't ask the system whether it's okay. It stalks, it tests, it finds the weak spot before anything else does.

That is the whole posture of this layer. Not a friendly assistant checking its own homework, but a predator pointed at your production AI on a schedule — paid to bring back real holes before a customer trips over one. The story is simple enough to repeat in a board meeting. The engineering underneath it is the hard part, and it comes down to a single question: *how do you pay a hunter so it brings back real prey — not easy prey, not imaginary prey?*

“Confidence is *not* correctness. An AI will hand you a wrong answer in the same self-assured tone it uses for a right one.”

THE PROBLEM, IN ONE LINE

01 The problem: a metric is a target

Goodhart's law, stated for agents.

The moment you reward an AI for finding problems, you have handed it a reason to manufacture them. This is not a hypothetical failure mode — it is the default one. A model optimized to “surface issues” will surface issues whether or not they exist, because the reward is attached to the act of flagging, not to the flag being correct.

Anyone who has managed a team to a metric knows exactly how this ends. Reward call volume and you get short, useless calls. Reward tickets closed and you get tickets closed prematurely. Reward findings and you get a loud auditor that fabricates drama to look useful. The auditor is behaving rationally; the *incentive* is broken.

So the central engineering task is not “get the AI to find things.” It is to build a scoring function where the *expected* value of bluffing is negative — where the rational strategy for a self-interested agent is, unavoidably, to tell the truth and cite it.

02 Four rules that make honesty the winning move

Reward engineering, in this system, reduces to four rules. Each one closes a specific avenue for gaming the score.

- ▶ **Paid to be right.** A finding earns points only if it is real and provable, and the points scale to how serious the problem is. Severity is not self-declared — it is inferred from where in the system the issue sits and what it touches.
- ▶ **Punished for bluffing.** If an auditor makes something up and a rival catches it, that finding scores worse than saying nothing at all. Silence is zero. A caught fabrication is negative. This single asymmetry is what removes the upside of crying wolf.
- ▶ **Cited, or it doesn't count.** Every finding must point to the exact place the problem lives — a node, a field, a step. A vague accusation with no location earns nothing, regardless of how plausible it sounds.
- ▶ **Never grades itself.** No model is ever allowed to score its own work, or the work of another model from the same lab. A rival platform always does the checking.

“A hunter paid by the kill will invent prey — so we only pay for prey we can **reproduce.”**

DESIGN PRINCIPLE · TIGER TEAM

WHY THE ASYMMETRY MATTERS MOST

If a false finding scored **zero** instead of negative, bluffing would be free — an auditor could spray speculative flags at no cost and occasionally get lucky. Making fabrication cost **less than silence** means the only safe move, when unsure, is to stay quiet. That is exactly the behavior you want from an inspector.

03 The scoring function

Each finding f raised by an auditor is scored independently by rival platforms and then resolved to a single number. The shape below is intentionally simple — legible incentives beat clever ones.

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score(f) = severity × consensus × citation − bluff_penalty
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severity $\in [0,1]$ — how much of the system the issue touches (revenue > cosmetic)
 consensus $\in [0,1]$ — fraction of independent rivals that confirmed it
 citation $\in \{0,1\}$ — hard gate: no exact location cited $\rightarrow$ the whole term is 0
 bluff_penalty > 0 — applied when a rival refutes an uncited or unsupported claim

- **Citation is a multiplicative gate** — an uncited finding collapses to zero no matter how severe it claims to be.
- **bluff_penalty is subtracted** — a refuted fabrication lands below zero, worse than staying silent.
- **Consensus is a factor** — a lone flag is worth a fraction of one two rivals confirm. Cross-lab agreement is the strongest signal.

Worked examples

Three findings from a single run, scored the same way. Note how the fabricated one nets negative.

FINDING	SEV.	CONSENSUS	CITED?	PENALTY	SCORE
Missing failure branch on a booking node (revenue-touching)	0.95	3 / 3 $\rightarrow$ 1.00	yes	0	+0.95
Two agents racing to write the same CRM field	0.80	2 / 3 $\rightarrow$ 0.67	yes	0	+0.54
Invented conflict on a “phone” field, unsupported by schema	0.60	0 / 3 $\rightarrow$ 0.00	no	0.35	-0.35

READ THE THIRD ROW CAREFULLY

The fabricated finding claimed real severity (0.60) — a naive scorer would have rewarded it. The citation gate zeroes the positive term, no rival confirmed it, and the bluff penalty pushes the total to **-0.35**: the auditor would have scored higher by staying silent. That is the entire point.

04 Why rivalry is load-bearing

The scoring function only works if the confirmations are independent. That is why the auditors are three different models from three different labs — Anthropic, OpenAI, and Google — each handed the same source of truth with no knowledge of what the others found.

“Different copies of the same AI share the same blind spots. Different **platforms don’t.”**

CORE ASSUMPTION

Using copies of one model, or three models from one lab, breaks the consensus term: they fail in the same places, in the same way, and they quietly flatter each other’s outputs. Rivalry across labs is what makes a “2 of 3 agreed” signal actually mean something.

Independent blind spots

Three labs fail in different places, covering far more ground than any one model.

No self-flattery

A model favors its own answers — but it can’t favor a rival’s.

No single point of failure

If one provider has a bad month, two independent checks still run.

No vendor lock-in

Confidence never rests on one company policing its own work.

05 What it produces, and why it matters

Every monthly run resolves to three artifacts, produced on a schedule without anyone asking:

- ▶ **A health score** — a single go / no-go number backed by ten specific checks. Below the line means “fix before you ship.”
- ▶ **A ranked list of real, provable issues** — the ones two or more rivals independently confirmed are flagged first, each pointing at exactly where the problem lives.
- ▶ **A ready-to-run fix** — not just “here’s what’s wrong,” but the exact repair instructions, reviewed before anything changes.

If AI is answering your phones, qualifying leads, or moving data between systems, it is already touching revenue. The gap between a business that trusts that quietly and one that verifies it on a schedule — with receipts — is the gap between “we think it’s working” and “we know it is.” That is the whole point of the Tiger Team: not to make the AI smarter, but to make it **accountable**.

The Tiger hunts on a schedule. The Sentinel never sleeps. — companion paper: “Standing Watch.”